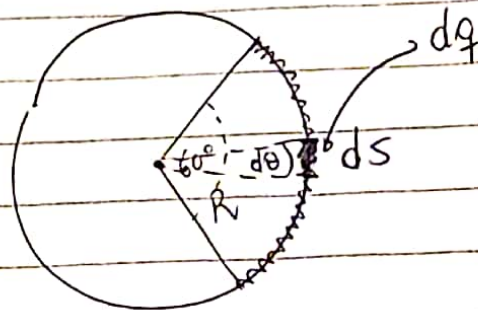


3)

a)



$$\lambda = \frac{dq}{ds}$$

$$ds = R d\theta$$

$$dq = \lambda ds$$

$$= \lambda R d\theta$$

$$Q = \int_0^{\pi/3} \lambda R d\theta$$

$$= -5 \left[\frac{nC}{m} \right] \cdot 0,2 \int_0^{\pi/3} d\theta$$

$$= -1,047 [nC]$$

$$b) \quad Q = \int_0^{2\pi} \lambda R d\theta \Rightarrow Q = -5 \left[\frac{nC}{m} \right] \cdot 0,2 \int_0^{2\pi} d\theta$$

$$= -6,283 [nC]$$